

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 03



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TOPICS TO BE COVERED

01 Binomial Coefficient

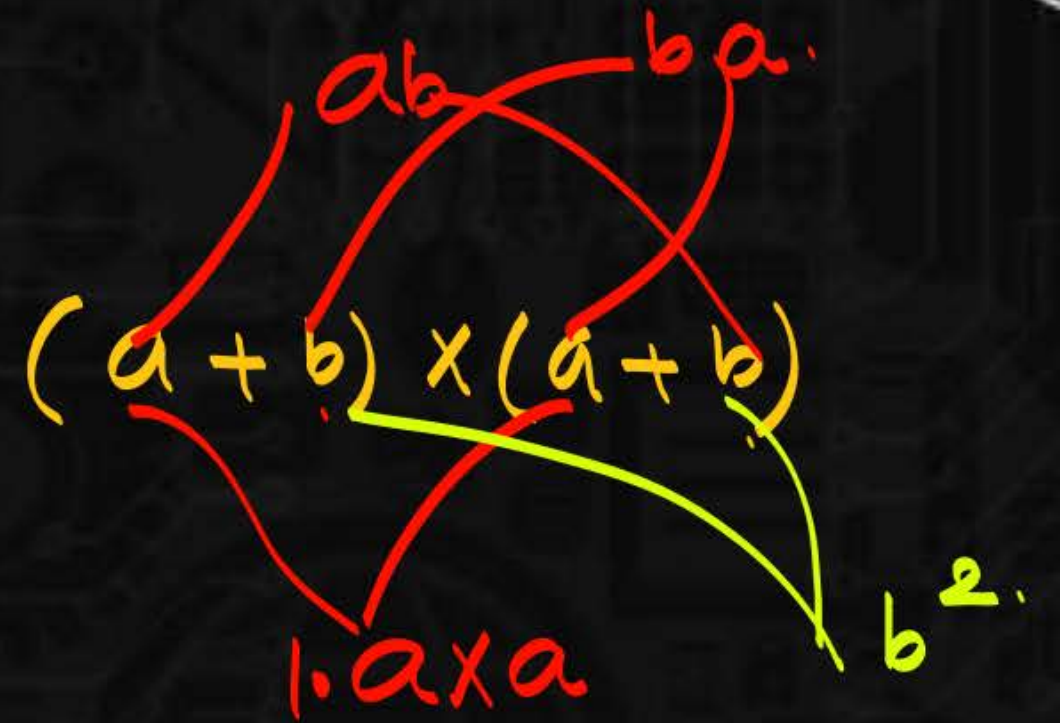
02 Problems in Binomial Coefficient

03 Extended Binomial q Coefficient

COMBINATORICS



$$(a+b)^2 = a^2 + 2ab + b^2.$$



$$(a+b) \times (a+b) \overset{\substack{\uparrow \\ \text{Sum Rule}}}{=} \underset{\substack{\uparrow \\ \text{P.R.}}}{1} \cdot a^2 + \underset{\substack{\uparrow \\ \text{Sum Rule.}}}{2} \cdot ab + \underset{\substack{\uparrow \\ \text{Way}}}{1} \cdot b^2$$

COMBINATORICS



$$(a+b)^3 = 1 \cdot a^3 + \underline{3a^2b} + 3ab^2 + b^3$$

$$(a+b) \times (a+b) \times (a+b)$$

$$3a^2b$$

$$\textcircled{3}a^2b$$

$$\boxed{a+} \times \boxed{a+} \times \boxed{+b}$$

out of 3
boxes
we want 1 b.
We can do
this
 3C_1 .

COMBINATORICS



$$(a+b)^3 = 1 \cdot a^3 + \underline{3a^2b} + 3ab^2 + b^3$$

$$= {}^3C_0 a^3 b^0 + {}^3C_1 a^2 b^1 + {}^3C_2 a^1 b^2 + {}^3C_3 a^0 b^3$$

1. power of expansion is same.

2. power of $a \downarrow$ power of $b \uparrow$

COMBINATORICS



$$(a+b)^n = n_{c0} a^n b^0 + n_{c1} a^{n-1} b^1 + n_{c2} a^{n-2} b^2 \dots n_{cn} a^0 b^n$$

$$(a+b)^n = \sum_{i=0}^n n_{ci} a^{n-i} b^i$$

$\rightarrow$ binomial coefficient

COMBINATORICS



$$(a+b)^n = \sum_{i=0}^n nC_i a^{n-i} b^i$$

$$a=1 \quad b=1.$$

$$(1+1)^n = \sum_{i=0}^n nC_i (1)^{n-i} (1)^i$$

$$2^n = \sum_{i=0}^n nC_i = nC_0 + nC_1 + nC_2 + nC_3 \dots nC_n.$$

$$nC_0 + nC_1 + nC_2 \dots nC_n = 2^n.$$

$$a=1 \quad b=-1.$$
$$(1-1)^n = \sum_{i=0}^n nC_i (1)^{n-i} (-1)^i$$

$$0 = \sum_{i=0}^n nC_i \cdot 1 \cdot (-1)^i$$

$$0 = nC_0 - nC_1 + nC_2 - nC_3$$

$$nC_1 + nC_3 \dots = nC_0 + nC_2.$$

COMBINATORICS



$$(a+b)^n = \sum_{i=0}^n {}^nC_i a^{n-i} b^i$$

$$a=1 \quad b=2.$$

$$(1+2)^n = \sum_{i=0}^n {}^nC_i \times (1)^{n-i} \times 2^i$$

$$\sum_{i=0}^n {}^nC_i \times 2^i = 3^n$$

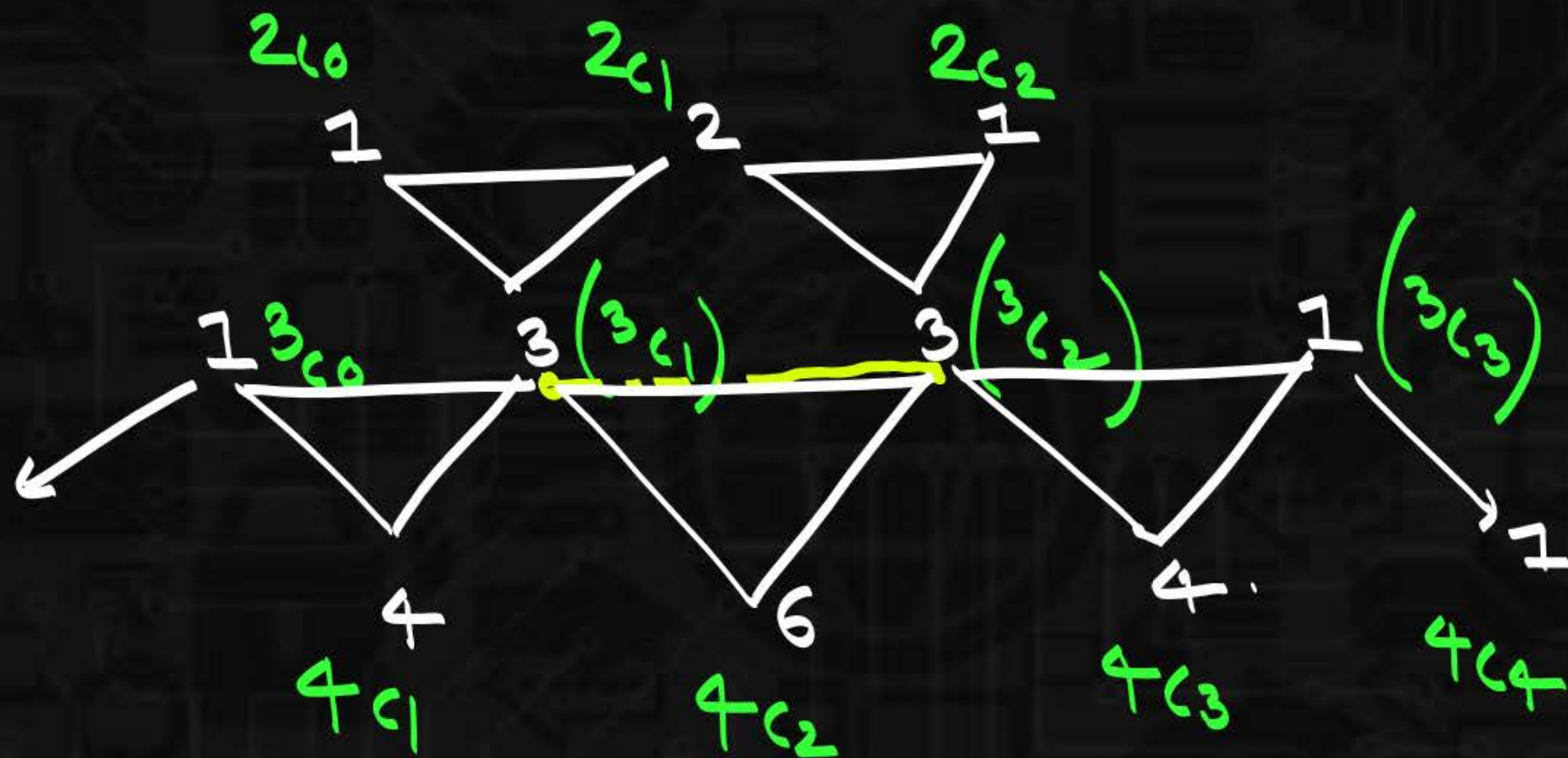
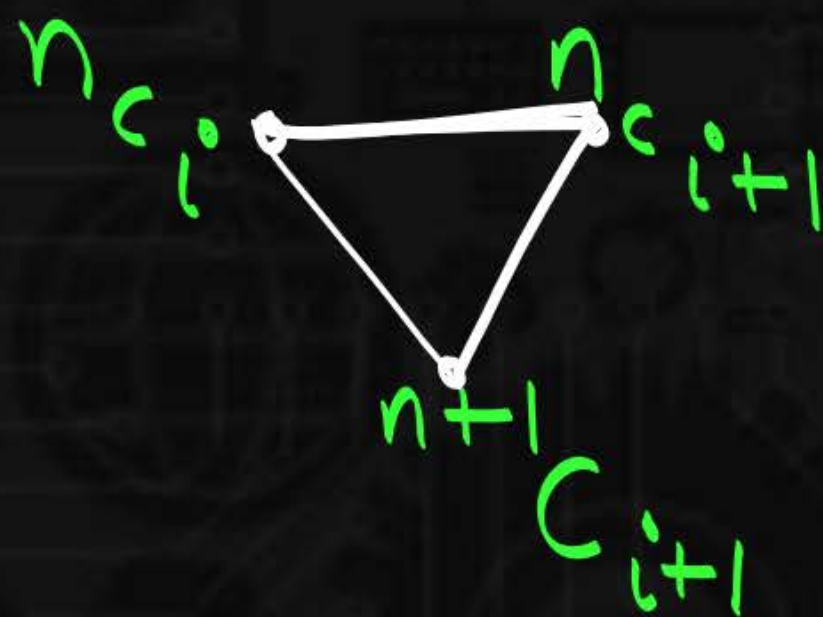
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COMBINATORICS



$$3c_1 + 3c_2 = 4c_2$$

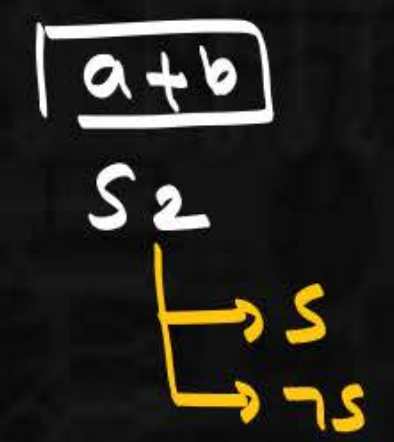
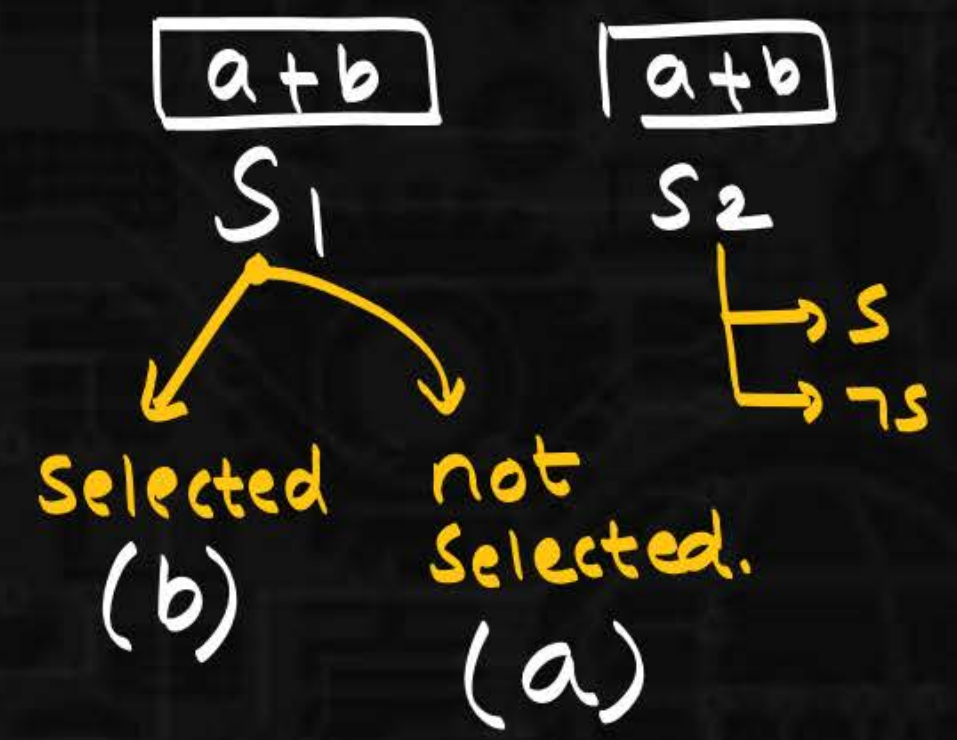
$$n_{c_i} + n_{c_{i+1}} = n_{c_{i+1}}$$



COMBINATORICS

how many ways we can select 6 students in a class of 10.

Ans $^{10}C_6$



Selected = b
not selected = a.

COMBINATORICS



$$(a+b) \times (a+b) \times (a+b) \dots (a+b)$$

Diagram showing the expansion of the product. A green box highlights the first three factors, and a green arrow points from the b in the first factor to the b in the second factor, illustrating the selection of b from each factor to form b^6 .

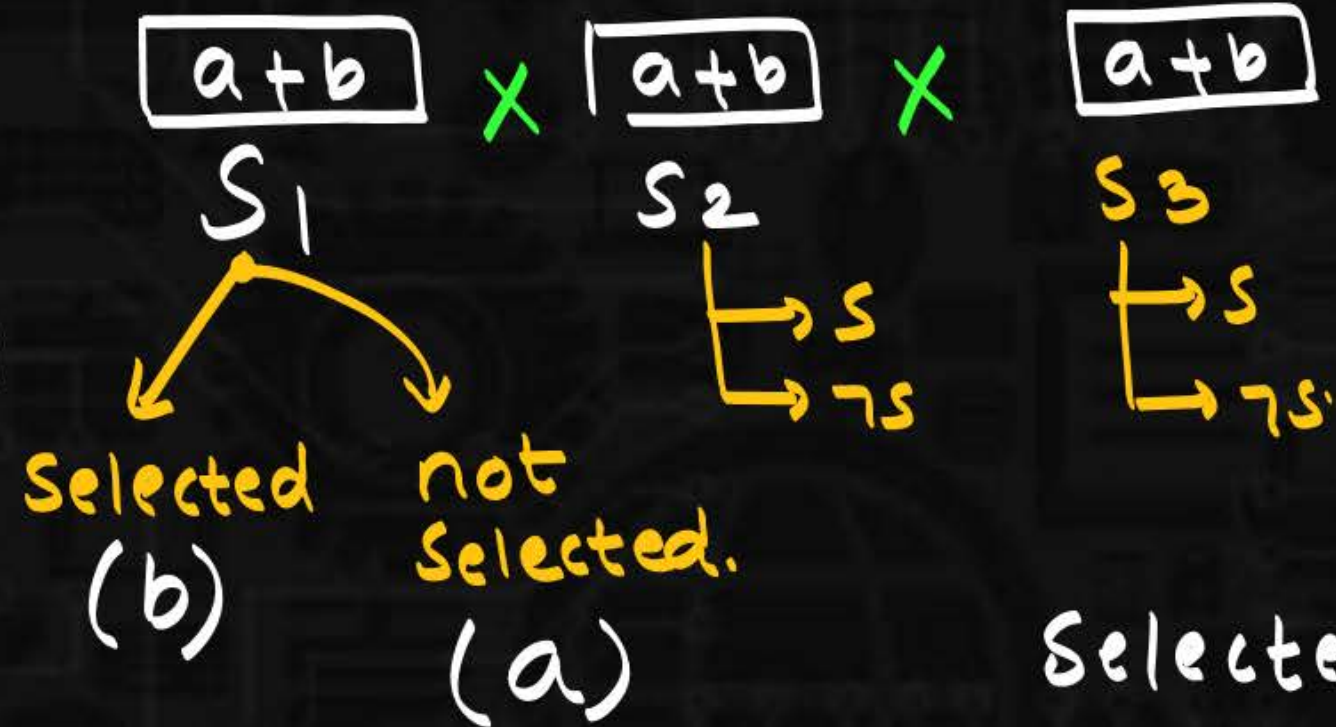
$$= (a+b)^{10}$$

Selecting 6 students = finding coefficient of b^6

$$(a+b)^n = \sum_{i=0}^n {}^nC_i (a)^{n-i} b^i$$

$$n=10$$

$$b^6 \quad i=6$$



Selected = b
not selected = a

$${}^nC_i a^{n-i} b^i \quad n=10 \quad i=6$$

$${}^{10}C_6 a^4 b^6$$

COMBINATORICS



find coefficient of x^{20}

$$(x+y)^{40} \quad n=40$$

$$n=40$$

$$i=20$$

$$x^{20} y^i$$

$$n C_i \cdot x^{n-i} y^i$$

$$40 C_{20}$$

$$20+i=40$$

$$i=20$$

$$(x+y)^n = \sum_{i=0}^n n C_i (x)^{n-i} y^i$$

coefficient x^{15}

$$(2x+3y)^{60} \quad n=60$$

$$60 C_{45} (2x)^{15} (3y)^{45}$$

$$x^{15} = x^{n-i}$$

$$15 = n-i$$

$$15 = 60-i$$

$$60 C_{45} \cdot 2^{15} \cdot x^{15} \cdot 3^{45} \cdot y^{45} \quad i=45$$

$$60 C_{45} \times 2^{15} \times 3^{45} = x^{15} y^{45}$$

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$$n=60 \quad i=45$$

$$\left(\frac{a}{2x} + \frac{b}{3y}\right)^{60}$$

$$a=2x \quad b=3y$$

$${}^nC_i (a)^{n-i} \times (b)^i$$

$${}^{60}C_{45} (2x)^{60-45} (3y)^{45}$$

$${}^{60}C_{45} (2x)^{15} (3y)^{45}$$

$${}^{60}C_{45} 2^{15} x^{15} 3^{45} y^{45}$$

$$\boxed{{}^{60}C_{45} 2^{15} 3^{45}} (x^{15} y^{45})$$

coefficient

coefficient
of x^{15}

$$\left(2x - \frac{3y}{4}\right)^{75}$$

$$2x + \left(-\frac{3y}{4}\right)$$

COMBINATORICS



$$n C_k = \frac{n!}{k! \times (n-k)!} = \frac{n(n-1)(n-2)\dots(n-k+1)\cancel{(n-k)!}}{k! \cancel{(n-k)!}}$$

$$= \frac{n(n-1)(n-2)\dots(n-k+1)}{k!}$$

$${}^{11}C_3 = \frac{(11)!}{3! (11-3)!} = \frac{11!}{3! \cdot 8!}$$

$$= \frac{11 \cdot 10 \cdot 9 \cdot \cancel{8!}}{3! \cdot \cancel{8!}} = \frac{11 \cdot 10 \cdot 9}{3!} = \frac{11(11-1)(11-2)}{3!(k)} = \frac{11 \times (11-1)(11-3+1)}{3!}$$

COMBINATORICS



$$n C_k = \frac{n(n-1)(n-2)\dots(n-k+1)}{k!} \text{ (binomial coefficient)}$$

Extended binomial coefficient:

$$-n C_k = \frac{(-n)(-n-1)(-n-2)\dots(-n-k+1)}{k!}$$

$$-n C_k = \frac{(-n-k+1)\dots(-n-2)(-n-1)(-n)}{k!}$$

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$$-n C_k = \frac{(-n-k+1) \dots (-n-2) (-n-1) (-n)}{k!}$$

take -1 common.

$$-n C_k = \frac{(-1)(n+k-1) \dots (-1)(n+2) (-1)(n+1) (-1)(n)}{k!}$$

$$-n C_k = (-1)^k \frac{(n+k-1) \dots (n+2)(n+1)(n)}{k!}$$

$$-n C_k = (-1)^k \frac{(n+k-1)!}{k! \times (n-1)!}$$

$$\times \frac{(n-1)!}{(n-1)!}$$

COMBINATORICS



$$-nC_k = (-1)^k \times \frac{(n+k-1)!}{k! \times (n-1)!}$$

$$-nC_k = (-1)^k \times {}^{n+k-1}C_k$$

$${}^{n+k-1}C_k$$

$$= \frac{(n+k-1)!}{k! \times (n+k-1-k)!}$$

$$-nC_k = \frac{(n+k-1)!}{k! \times (n-1)!}$$

COMBINATORICS



$$-nC_k = (-1)^k n+k-1 c_k.$$

$$-11c_3 = -nC_k.$$

$$n=11 \quad k=3$$

$$(-1)^k n+k-1 c_k$$

$$(-1)^3 11+3-1 c_3 = (-1)^3 13 c_3 = (-1)^3 c_3$$

$$-11c_3 = (-1) \times 13 c_3$$

k odd -ve

k even +ve

$$-12c_4 = -nC_k.$$

$$n=12 \quad k=4$$

$$(-1)^k n+k-1 c_k.$$

$$(-1)^4 \times 12+4-1 c_4.$$

$$\downarrow$$
$$1 \times 15 c_4$$

$$-12c_4 = 15c_4.$$

COMBINATORICS



$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{1-x} = (1-x)^{-1} = (1+(-x))^{-1}$$

$$(a+b)^n = nC_0 a^{n-0} b^0 + nC_1 a^{n-1} b^1 + \dots$$

$$\left(1 + \frac{(-x)}{1}\right)^{-1} = -1C_0 (1)^{-1-0} (-x)^0 + -1C_1 (1)^{-1-1} (-x)^1 + \dots$$

$\downarrow \quad \downarrow \quad \downarrow$
 $1 \quad 1 \quad 1$

$$-1C_0 = -nC_k = (-1)^k n+1-k C_k$$

$$k=0 \quad n=1 \quad = 1^{-1-0} C_0$$

$$= 1 \cdot 0 C_0 = 1$$

COMBINATORICS



$$\frac{1}{1-x} = (1-x)^{-1} = (1+(-x))^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$(a+b)^n = nC_0 a^{n-0} b^0 + nC_1 a^{n-1} b^1 + nC_2 a^{n-2} b^2 + \dots$$

$$\frac{(1+(-x))^{-1}}{a^0 b^0} = -1C_0 \binom{-1}{0} (-x)^0 + -1C_1 \binom{-1}{1} (-x)^1 + \underbrace{-1C_2 \binom{-1}{2} (-x)^2}_{+ve} + \underbrace{-1C_3 \binom{-1}{3} (-x)^3}_{-ve} + \dots$$

$n = -1$

$$(-1)^k n+k-1C_k$$

$$(-1)^2 1+2-1C_2$$

$$= 1 \cdot 2C_2 = 1$$

COMBINATORICS



Q3

Up to 5 term

$$\frac{1}{1-an} =$$

$$\frac{1}{1+an}$$

